

Project 3: SQL Data Analysis

DecodeLabs Data Analytics Internship — Batch 2026 Submitted by: Zunesha Naeem (Zoha) Registration No: BAN243007 BS Business Analytics, Capital University of Science and Technology (CUST)

Project Overview

This project applies core SQL skills — SELECT, WHERE, GROUP BY, HAVING, ORDER BY, and aggregate functions (COUNT, SUM, AVG) — to extract business insights from an e-commerce orders dataset containing 1,200 records.

Dataset

The dataset (Cleaned_Dataset_for_Data_Analytics.csv) contains the following columns:

Column	Description
OrderID	Unique identifier for each order
OrderDate	Date the order was placed
CustomerID	Unique identifier for each customer
Product	Product purchased
Quantity	Number of units ordered
UnitPrice	Price per unit (USD)
ShippingAddress	Delivery address
PaymentMethod	Payment mode used
OrderStatus	Order status (Delivered, Shipped, Pending, Cancelled, Returned)
TrackingNumber	Shipment tracking number
ItemsInCart	Number of items in customer's cart
CouponCode	Coupon applied (or "No Coupon")
ReferralSource	Marketing channel that referred the customer
TotalPrice	Total order value (Quantity × UnitPrice)

Files in this Repository

- **Orders_Full_AllInOne.sql** — Table schema, full dataset (1,200 rows), and sample query. Can be run end-to-end in any SQLite playground (e.g., DB Fiddle).
- **Project3_SQL_Data_Analysis.docx** — Full report containing all 11 queries, results, and business interpretations.

- **Cleaned_Dataset_for_Data_Analytics.csv** — Cleaned and formatted dataset used for analysis.

Queries Covered

1. High-value orders (SELECT, WHERE, ORDER BY)
2. Order status distribution (GROUP BY, COUNT)
3. Total revenue by product (GROUP BY, SUM, ORDER BY)
4. Average order value by payment method (GROUP BY, AVG, COUNT)
5. Delivered orders in 2024 by product (multi-condition WHERE, GROUP BY)
6. High-performing products (GROUP BY, HAVING)
7. Percentage contribution of each product to total revenue (subquery)
8. Marketing channel performance (GROUP BY, COUNT, SUM, AVG)
9. Top 5 customers by total spend (GROUP BY, ORDER BY, LIMIT)
10. Coupon usage vs. average order value (GROUP BY, AVG, COUNT)
11. Cancellation/return rate by product (CASE, GROUP BY, percentage)

Key Insights

- Revenue is well diversified across product categories — Chair and Printer each contribute ~15.5% of total revenue, while Phone contributes the least (~12%).
- Cancelled and Returned orders make up over 41% of all orders, with a consistent 39%–44% issue rate across every product — suggesting a systemic fulfilment issue.
- Credit Card customers have the highest average order value (\$1,127.55); Instagram is the top marketing channel by both order volume and revenue.
- The WINTER15 coupon is linked to the lowest average order value, even below orders with no coupon at all.
- Most top-spending customers reached their totals via a single large order rather than repeat purchases.

How to Run

1. Open `Orders_Full_AllInOne.sql`.
2. Copy the entire contents into an SQLite playground (e.g., [DB Fiddle](#)).
3. Click “Run All” — this creates the Orders table, loads all 1,200 rows, and runs a sample query.
4. Replace the final SELECT statement with any of the 11 queries to reproduce results from the report.